

Eng. Youstina Salah Nathan

Full-Stack AI/ML Engineer | Data Analyst & Data Scientist | Generative AI (LLMs, RAG, MLOps)

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PROFESSIONAL SUMMARY

Full-stack AI engineer and data scientist experienced in building end-to-end intelligent systems — from Python/Flask and ASP.NET Core backends to production-ready ML pipelines and LLM-powered features. Shipped an AI Interview System (OpenCV, MediaPipe, Whisper, Claude & Gemini APIs), a citation-traceable RAG chatbot (Gemini embeddings, Supabase vector search), and a CV Analyzer using TF-IDF/Cosine/Jaccard similarity, plus three end-to-end predictive modeling projects (scikit-learn, Pandas, NumPy) through a CodeAlpha ML internship. Comfortable across the stack: REST API design & auth, SQL/NoSQL data layers, model evaluation, prompt engineering, and data visualization (Power BI, advanced SQL) to turn model output into decisions. IT graduate certified by Microsoft, Udacity, Oracle, Cisco, Route IT, and ITIDA/NTI. Trilingual, adaptable, and quick to learn new tools and systems — seeking a Full-Stack AI Engineer / Data Analyst / Data Scientist role.

TECHNICAL SKILLS

- Programming & Core: Python (professional), C#, C++, JavaScript, jQuery, HTML, CSS, SQL, OOP, Git, Java
- .NET & Backend: ASP.NET Core (Web API & MVC), Entity Framework Core, SQL Server, RESTful API Design, Auth, Flask, Model Fallback Chains
- AI / Machine Learning: Machine Learning, NLP, Computer Vision (OpenCV, MediaPipe), scikit-learn, TF-IDF & Cosine/Jaccard Similarity, Model Evaluation
- Generative AI & APIs: Google Gemini API, Anthropic Claude API, OpenAI Whisper, REST API Design, Prompt Engineering
- Data Analysis & BI: Advanced Excel, KPI Reporting & Dashboards, Power BI, Data Validation & Consolidation, Data Visualization, Statistical Analysis, Pandas, NumPy, Big Data Analysis, ETL, DAX
- Databases: Excel (advanced), SQLite, Oracle Database, MySQL, SQL Server, DB Design & Normalization, Oracle Cloud
- Doc/Data Processing: pdfplumber, python-docx, ReportLab
- Tools: Jupyter, MS 365, Dynamics 365 CRM, Cisco Packet Tracer, Visual Studio

EXPERIENCE & PROJECTS

AI/Python Backend Developer — Jobify (AI-Powered Employment Platform, Graduation Project, EELU)2025 – 2026

- Owned the AI/Python backend across a 3-role team, delivering an AI Interview System (OpenCV, MediaPipe, Whisper, Claude & Gemini APIs), a Flask-based CV Generator with model fallback chains, and a CV Analyzer (TF-IDF, Cosine & Jaccard Similarity).
- Presented full technical defense of the system architecture to academic and technical stakeholders (Dr. Mahmoud Bassiouni, Eng. Nourhan Khashaba).

Backend Developer — E-Commerce Web Application — Route IT Training Center, .NET Core Diploma Project
2025

- Built a full-stack ASP.NET Core app (MVC storefront + Web API) with EF Core/SQL Server data layer, user authentication & role-based access, product catalog, cart, and checkout flow.

Big Data Analysis Trainee — ITIDA & NTI (Summer Training Program) Jul – Aug 2025

- Completed a 120-hour Big Data Analysis program (90 technical + 30 freelancing hours) covering large-scale data collection, consolidation, processing & analysis; scored 87.5%, certified jointly by ITIDA & NTI.

Machine Learning Intern Project — CodeAlpha (Remote) 2024 – 2025

- Delivered 3 end-to-end ML projects (Iris Classification, Unemployment Rate Analysis, Car Price Prediction) as production-ready notebooks using Pandas, NumPy, scikit-learn; documented outcomes via LinkedIn write-ups.

Data Analysis Capstone Project — Breast Cancer Health Analysis Dashboard — Udacity 2024

- Cleaned and validated 50,000+ records (Pandas/NumPy), built a data model and interactive Power BI dashboard with 8+ KPI visualizations, and wrote 20+ advanced SQL queries (JOINS, CTEs, window functions) to support performance reporting.
- Built an interactive Power BI dashboard (Excel, Power BI) applying data visualization best practices to breast cancer health data, surfacing diagnosis trends, risk factors, and key patient metrics to support early-detection insights.

- SQL Data Analysis Project — Seed & Plant Growth Analysis Dashboard** — Udacity 2024
- Designed complex multi-table SQL queries (subqueries, GROUP BY, CASE logic) and performed 12-month trend analysis, identifying top-performing segments and flagging variances in revenue patterns for review.
 - Designed an Excel and Power BI-style dashboard (Excel, Power BI) modeling seed germination and plant growth data to track growth-rate metrics and prepared presentation-ready charts and summaries for review.
- Network Infrastructure Design** — Cisco Networking Academy 2023
- Designed and simulated a small business network in Cisco Packet Tracer applying TCP/IP, subnetting, and VLAN configuration across 3 simulated subnets.
- Database Management System (Library System)** — EELU 2023
- Designed a normalized (3NF) Oracle Database schema; implemented stored procedures, triggers, and 15+ queries managing 1,000+ records within a 2-week sprint.
- Train Ticket Reservation System — C++ (Academic Project)** — Microsoft Student Club-EELU 2024
- Built a C++ console-based reservation system handling booking, cancellation, and seat/schedule management using core data structures and file handling.
- PostpartumRAG — Maternal Mental Health RAG Chatbot** — Orange Digital Center, ITIDA, TIEC, Creativa & Instant Software Solutions Hackathon Aug 2026
- Built a grounded, citation-traceable RAG chatbot (Angular, Express.js, MongoDB, Python microservice, Gemini embeddings, Supabase vector search) to support maternal mental health guidance, using section-aware chunking of WHO, mhGAP, and NICE CG192 clinical sources.
 - Placed among the Top 3 teams out of all competing teams; received a certificate of achievement from the hackathon organizers.
- AI Fundamentals with IBM SkillsBuild** — Cisco Networking Academy 2026
- Covered core AI concepts including machine learning basics, neural networks, natural language processing, computer vision, and responsible/ethical AI principles.

EDUCATION

B.Sc. Information Technology — Egyptian E-Learning University (EELU), Assiut, Egypt

Relevant Coursework: Data Structures, Database Management, Computer Networks, Programming Basics, Data Analysis & Interpretation.

CERTIFICATIONS

- Back-end (.NET Core) Diploma — Route IT Training Center (Nov 2025)
- Power BI Data Modelling Basics — Microsoft (Aug 2026)
- Data Analyst 101 — Microsoft (Aug 2026)
- Business Analytics with Excel — Microsoft (Aug 2026)
- Security, Compliance & Identity Fundamentals — Microsoft (2024)
- Big Data Analysis Training — ITIDA & NTI, 120 hrs, 87.5% (2025)
- Data Analysis Professional Track — Udacity (2024)
- SQL for Data Analysis — Udacity (2024)
- Oracle Cloud Infrastructure Foundations — Oracle (2023)
- AI Fundamentals with IBM SkillsBuild — Cisco Networking Academy (2026)
- Oracle Database Foundations — Oracle (2023)
- Device Management Professional — Microsoft (2023)
- Networking Essentials and Switching — Cisco Networking Academy (2023)
- English Language Online Course — The American University in Cairo (AUC) (2022)

VOLUNTEERING & EXTRACURRICULAR

- Team Member, Huawei Volunteer Program — collaborated on volunteering initiatives and community activities.
- Microsoft Student Club-EELU — developing teamwork and collaboration skills.

LANGUAGES

Arabic: Native | English: Professional Working Proficiency | French: Intermediate